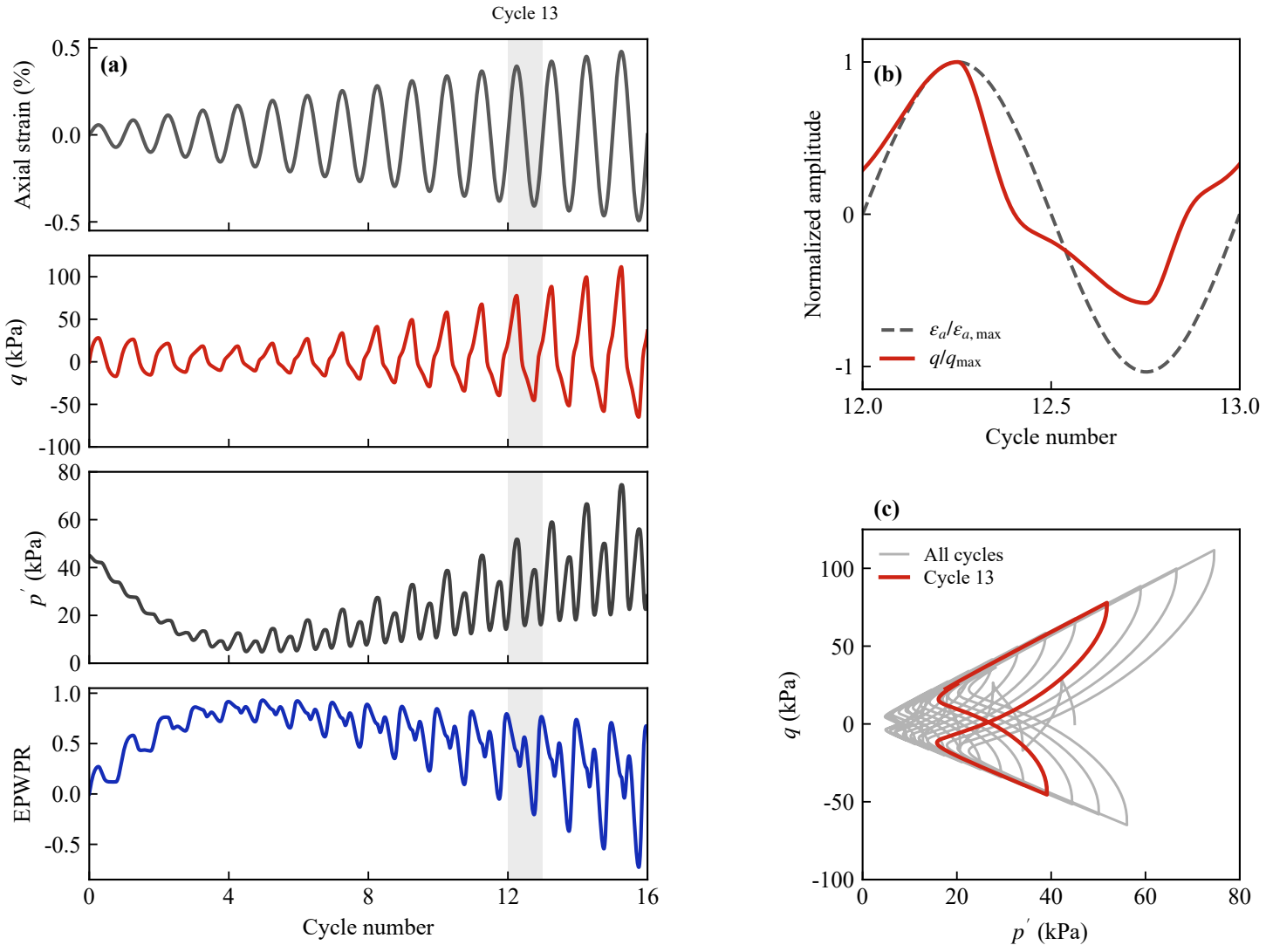

Cycle 13
(a)
Axial strain (%)
0.5
0.0
-0.5
q (kPa)
100
50
0
-50
-100
p′ (kPa)
80
60
40
20
0
EPWPR
1.0
0.5
0.0
-0.5
0
4
8
12
16
Cycle number
(b)
Normalized amplitude
1
0
-1
$\varepsilon_a/\varepsilon_{a,\max}$
$q/q_{\max}$
12.0
12.5
13.0
Cycle number
(c)
All cycles
Cycle 13
q (kPa)
100
50
0
-50
-100
0
20
40
60
80
p′ (kPa)